Regression Limitation

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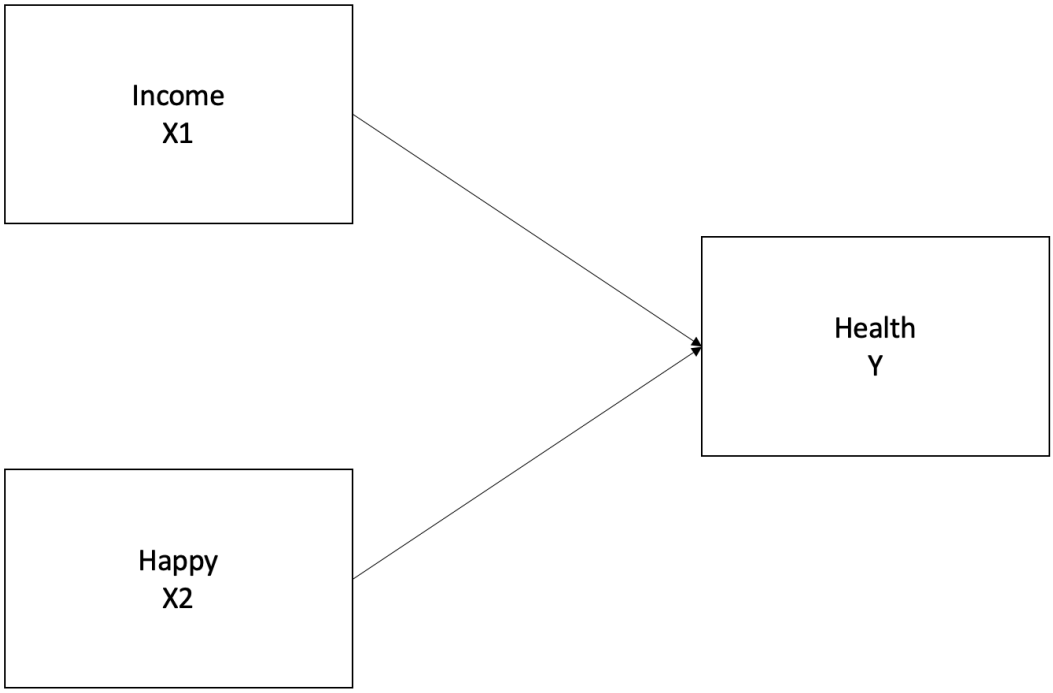
2025-05-01

Problem

We are interested in knowing if income and happy predict health. Please import the data (**health.sav**), draw the diagram, and run the analysis.

Solution



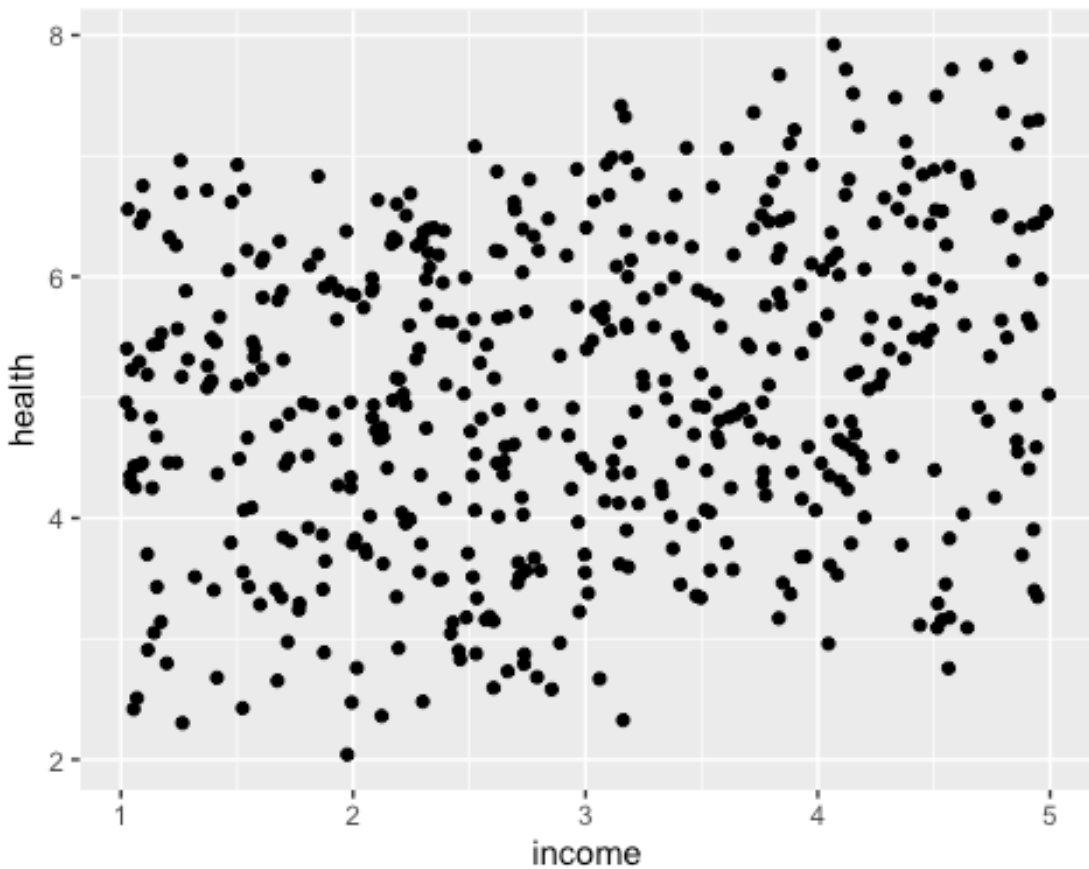

```
#load packages
library(tidyverse)
library(psych)
library(haven)

#import data

data<-read_sav("health.sav")

#take a look at our data
data
```

```
# A tibble: 500 × 3
   income happy health
    <dbl> <dbl>  <dbl>
 1   1.53  5.74   6.72
 2   2.55  5.59   4.82
 3   1.06  3.00   4.43
 4   3.48  3.85   3.36
 5   1.02  5.13   4.96
 6   3.51  6.85   4.92
 7   1.87  3.13   3.86
 8   4.48  6.15   5.79
 9   4.02  4.84   6.05
10   4.49  6.31   5.56
# ℹ 490 more rows
```

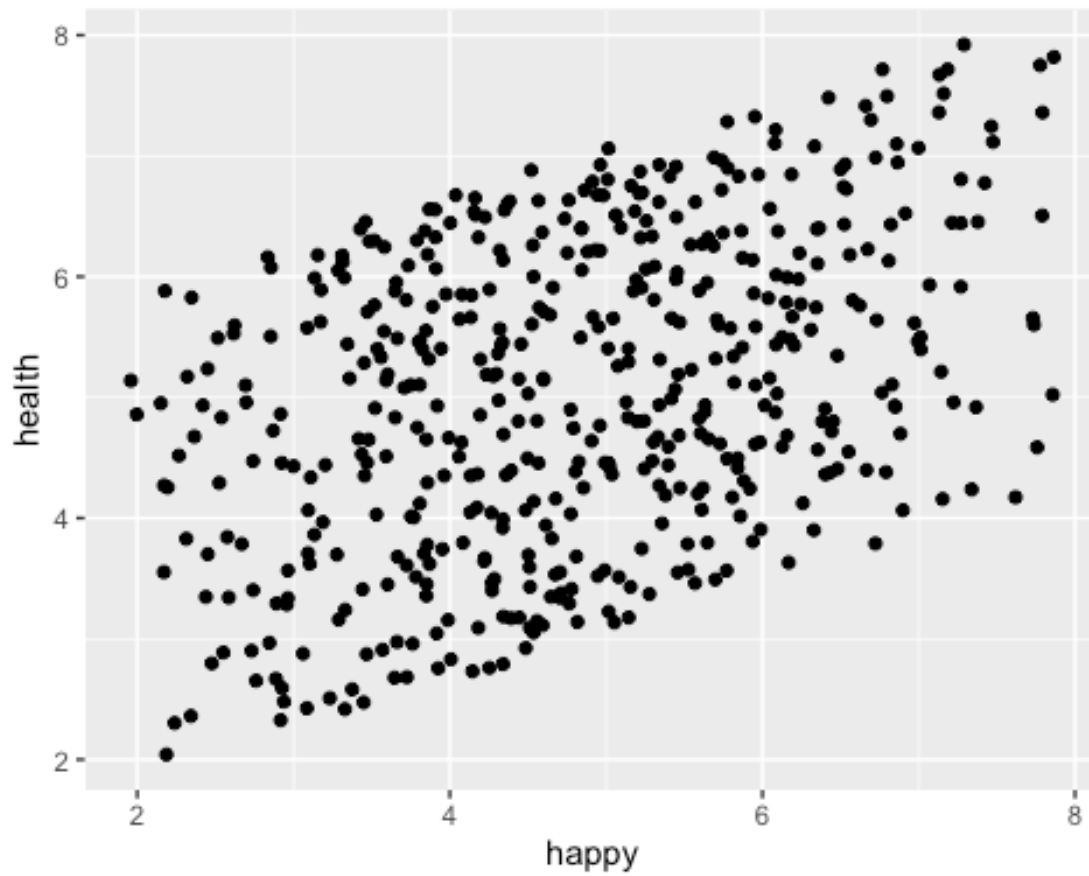
```
#descriptive stats
describe(data)
```

```
       vars   n mean   sd median trimmed  mad  min  max range  skew kurtosis
income    1 500 2.96 1.13   2.96    2.96 1.37 1.02 4.99  3.97  0.03    -1.13
happy     2 500 4.82 1.36   4.78    4.81 1.48 1.96 7.86  5.90  0.09    -0.72
health    3 500 5.05 1.28   5.10    5.07 1.50 2.04 7.92  5.88 -0.10    -0.83
         se
income 0.05
happy  0.06
health 0.06
```

```
#visualize our data

data |>
  ggplot(mapping = aes (x = income, y = health)) +
  geom_point()
```




```
data |>
  ggplot(mapping = aes (x = happy, y = health)) +
  geom_point()
```



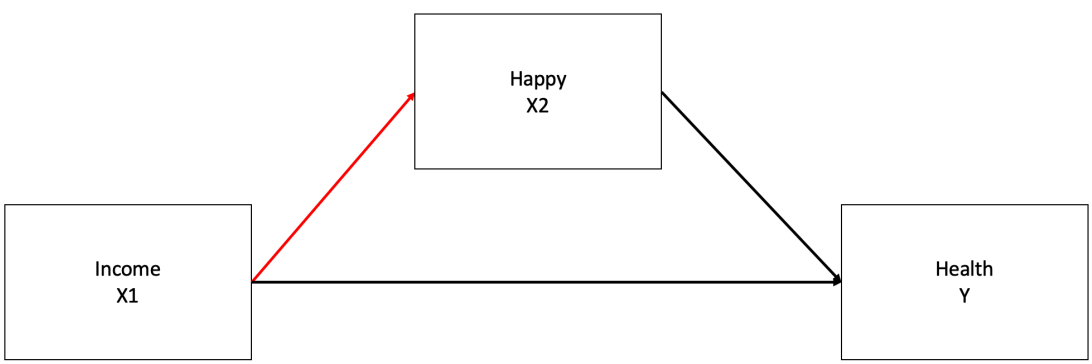

```
#model our data

fit<-lm(health ~ income + happy, data = data)
summary(fit)


Call:
lm(formula = health ~ income + happy, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.0732 -0.9668  0.0511  1.0244  1.9448 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 3.157912   0.198570  15.903   <2e-16 ***
income      0.001315   0.054063   0.024    0.981    
happy       0.392358   0.044938   8.731   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.165 on 497 degrees of freedom
```

```
Multiple R-squared:  0.1735,    Adjusted R-squared:  0.1702
F-statistic: 52.18 on 2 and 497 DF,  p-value: < 2.2e-16

#evaluation: overall model + predictor(s)

#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the
predictability of our model. The higher the better.

#interpretation: About 17.35% of variance in health is explained by income
AND happy.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Hoding happy to be equal, income does not significantly
predict health. One unit increase in income is expected to be associated with
an increase of 0.001315 units in health.

#interpretation: Hoding income to be equal, happy significantly and
positively predicts health. One unit increase in happy is expected to be
associated with an increase of 0.392358 units in health.
```

Discussion

Since income is not a significant predictor, we may think that income is not important for one's health. But is this an accurate conclusion?

What if there is sequencing between income and happy? Like this:



In this case, income predicts happy, which, in turn, predicts health. So, if you have higher income, you are more likely to be happy. If you are more likely to be happy, you are more likely to be healthy. Hence, we can say that although income is not directly related to health, it is indirectly related to health through happy. But how can we model this complex relation? In simple/multiple linear regression models, we are not able to model sequencing between between predictors (i.e., the temporal order). This is when we turn to the first

model we are going to learn within the framework of conditional process analysis: **the simple mediation model**.